Started on	Sunday, 24 July 2022, 5:55 PM
State	Finished
Completed on	Sunday, 24 July 2022, 5:55 PM
Time taken	7 secs
Marks	0.00/56.00
Grade	**0.00** out of 100.00

Question **1**

Not answered

Marked out of 1.00

Flag question

Which of the following can be called a major tranquilliser?

Select one:

- Risperidone
- Chlorpromazine
- Propofol
- Chlordiazepoxide
- Secobarbital

Explanation:

Historically, antipsychotics were known as **major tranquillisers**, while benzodiazepines and barbiturates were known as **minor tranquillisers**. **Delay and Deniker** developed the first antipsychotic agent, **chlorpromazine**, at the Rhône-Poulenc Pharmaceutical Company, under the direction of Paul **Charpentier**. First studied by a French military surgeon for trauma, psychiatrists soon adapted it for its '**tranquilising**' effects.

Ref: Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. p.3.

Westermeyer J. Cultural Psychiatry in the French-Speaking World. In Cultural Psychiatry 2013 (Vol. 33, pp. 56-63). Karger Publishers.

The correct answer is: Chlorpromazine

Question **2**

Not answered

Marked out of 1.00

Flag question

A 45-year-old man with paranoid schizophrenia is started on a thienobenzodiazepine. Which of the following drugs is prescribed here?

Select one:

- Paliperidone
- Piperazine
- Olanzapine
- Quetiapine
- Clozapine

Explanation:

Olanzapine is a thienobenzodiazepine with antagonism for multiple receptors. Thienobenzodiazepines are characterised as multi-receptor-targeted-acting-agents. Olanzapine is an atypical antipsychotic agent that is structurally related to clozapine.

Ref: Mendonça Júnior FJ et al. Benzo-and thienobenzo-diazepines: multi-target drugs for CNS disorders. Mini reviews in medicinal chemistry. 2015 Jun 1;15(8):630-47.

The correct answer is: Olanzapine

Question **3**

Not answered

Marked out of 1.00

Flag question

Select the reason why clozapine was initially withdrawn before Kane's landmark trial:

Select one:

- High costs
- Fatality due to agranulocytosis
- Lack of efficacy
- Unclear dosing schedule
- Excessive sedation

Explanation:

Clozapine is a dibenzodiazepine derivative. Shortly after its introduction to clinical practice in the mid-1970s, it was withdrawn because of several episodes of fatal agranulocytosis in patients on treatment. It was thought to have special efficacy in treatment-resistant schizophrenia, and this clinical belief was supported by an important trial by Kane et al. (1988), leading to its reintroduction in psychiatric practice, albeit with strict limitations to its prescription.

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 218.

The correct answer is: Fatality due to agranulocytosis

Question **4**

Not answered

Marked out of 1.00

Flag question

What is cyproterone acetate?

Select one:

- Glucocorticoid analogue
- Testosterone antagonist
- Mineralocorticoid antagonist
- Progesterone antagonist
- Testosterone partial agonist

Explanation:

Cyproterone acetate is an anti-androgen. There is evidence for its efficacy in the treatment of hypersexuality and paraphilias. It works by preventing testosterone from binding to the receptors

(antagonist).

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 507.

The correct answer is: Testosterone antagonist

Question **5**

Not answered

Marked out of 1.00

Flag question

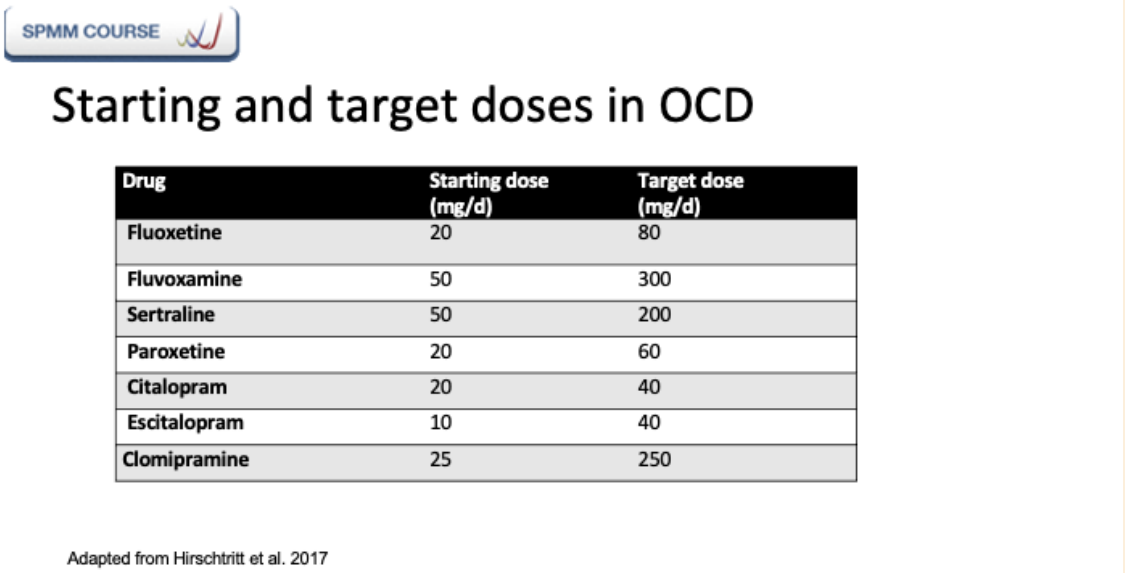
Which drug is useful for the treatment of OCD?

Select one:

- Trazadone
- Duloxetine
- Phenelzine
- Venlafaxine
- Sertraline

Explanation:

SSRIs should be considered first-line in the treatment of OCD (no clear superiority of any one agent, high doses usually needed, at least 12 weeks for treatment response, long-term therapy).



Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 386.

Hirschtritt ME et al. Obsessive-compulsive disorder: advances in diagnosis and treatment. Jama. 2017 Apr 4;317(13):1358-67.

The correct answer is: Sertraline

Question **6**

Not answered

Which of the following is true about placebo response in drug trials?

Marked out of 1.00

Flag question

Select one:

- No placebo response is seen in personality disorders
- A group of 'placebo reactors' with stereotyped personality can be identified
- Degree of placebo response is similar irrespective of the treatment studied
- Opioids may have a role to play in placebo response
- Placebo response in depression is gradual but persistent

Explanation:

Placebo effects have been shown to be associated with the release of substances such as endogenous opioids, endocannabinoids, dopamine, oxytocin, and vasopressin.

Ref: Colloca L, Barsky AJ. Placebo and nocebo effects. New England Journal of Medicine. 2020 Feb 6;382(6):554-61.

The correct answer is: Opioids may have a role to play in placebo response

Question **7**

Not answered

Marked out of 1.00

Flag question

The drug alprazolam belongs to which of the following class of psychotropics?

Select one:

- Stimulants
- Antipsychotics
- Anticholinesterases
- Benzodiazepines
- Antidepressants

Explanation:

The benzodiazepines derive their name from their molecular structure. They share a common effect on receptors that have been termed benzodiazepine receptors, which in turn modulate GABA activity. The absorption, the attainment of peak concentrations, and the onset of action are quickest for diazepam, lorazepam, alprazolam, triazolam, and estazolam.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 669.

The correct answer is: Benzodiazepines

Question **8**

Not answered

Marked out of 1.00

Flag question

Choose the correct match from the following pairs:

Select one:

- Risperidone: dibenzoxapine
- Droperidol: butyrophenone
- Aripiprazole: benzisothiazole
- Thioridazine: diphenyl butyl piperidine
- Flupentixol: dihydroindole

Explanation:

Haloperidol, droperidol, and benperidol are butyrophenones. They are high potency, with the potential for troublesome extra-pyramidal side effects.

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 209.

Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 229.

The correct answer is: Droperidol: butyrophenone

Question **9**

Not answered

Marked out of 1.00

Flag question

Which of the following drugs is correctly matched with its metabolite?

Select one:

- naloxone - naltrexone
- imipramine - nomifensine
- citalopram - escitalopram
- risperidone - paliperidone
- fluoxetine - duloxetine

Explanation:

Paliperidone is a benzisoxazole derivative and is the major active metabolite of risperidone.

Escitalopram is not a metabolite of citalopram but is comprised of only the pure S enantiomer.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 610.

Stahl SM. Stahl's Essential Psychopharmacology: Neuroscientific Basis and Practical Applications, 5th Edition. 2021. p. 296.

The correct answer is: risperidone - paliperidone

Question **10**

Not answered

Marked out of 1.00

Flag question

Discovery of tranquilising properties of chlorpromazine is associated with

Select one:

- Delay and Deniker
- Bleuler
- Cade
- Sen and Ghose
- Kane

Explanation:

Early in their Vietnam War (late 1940s and early 1950s), French pharmacologists employed a combination of the antihistamine phenergan and an opioid medication in the wounded, facilitating analgesia, alleviating fear, and opposing shock. While working on similar compounds, **Delay and**

Deniker developed the first antipsychotic agent, **chlorpromazine**, at the Rhône-Poulenc Pharmaceutical Company, under the direction of Paul **Charpentier**. First studied by a French military surgeon for trauma, psychiatrists soon adapted it for its '**tranquilising**' effects.

Ref: Westermeyer J. Cultural Psychiatry in the French-Speaking World. In Cultural Psychiatry 2013 (Vol. 33, pp. 56-63). Karger Publishers.

The correct answer is: Delay and Deniker

Question **11**

Not answered

Marked out of 1.00

Flag question

A 15-year-old boy has not been to his school for last few weeks as he has been disturbed by derogatory auditory hallucinations. He has been hearing voices in the third person for two months and is receiving messages from the television. The most likely treatment option is

Select one:

- Parental training
- Family therapy
- Fluoxetine
- Cognitive behavioural therapy for anxiety
- Low dose risperidone

Explanation:

Antipsychotic medication is first-line treatment for first-episode psychosis. Olanzapine, amisulpride, and risperidone may be marginally better than other options. Determine the choice of medication together with the patient, considering the adverse effect profile.

Ref: Barnes TR et al. Evidence-based guidelines for the pharmacological treatment of schizophrenia: Updated recommendations from the British Association for Psychopharmacology. Journal of Psychopharmacology. 2020 Jan;34(1):3-78.

The correct answer is: Low dose risperidone

Question **12**

Not answered

Marked out of 1.00

Flag question

Ethnic differences in psychiatric drug effects are noted in which of the following pharmacological actions?

Select one:

- Blood levels of haloperidol
- Hydroxylation of tricyclics
- Alcohol metabolism
- Prolactin response to antipsychotics
- All of the listed options

Maximal haloperidol concentration in plasma after rapid tranquillisation is significantly high for Asians than Caucasians. Nearly 40% Asians and around 60% South American Native Indians lack Aldehyde dehydrogenase enzyme in sufficient amounts to metabolise alcohol. Caucasians appear to have lower plasma levels of tricyclic antidepressants and attain plasma peaks later when compared with Asians (of Far Eastern ancestry as well as those from the Indian subcontinent). These differences

have been attributed to a greater incidence of slow hydroxylation among Asians when compared with Caucasians. . On administration of antipsychotics, Asian subjects were reported to produce greater serum prolactin levels than Caucasian subjects. (Bhugra, D., & Bhui, K. (1999). Ethnic and cultural factors in psychopharmacology. Advances in Psychiatric Treatment, 5, 89-95.)

The correct answer is: All of the listed options

Question **13**

Not answered

Marked out of 1.00

Flag question

Which of the following is a substituted benzamide used as an antipsychotic?

Select one:

- Amisulpride
- Benzhedrine
- Paliperidone
- Risperidone
- Olanzapine

Explanation:

Sulpiride and amisulpride are substituted benzamides that are used as antipsychotics.

Ref: Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. p. 3.

The correct answer is: Amisulpride

Question **14**

Not answered

Marked out of 1.00

Flag question

Discovery of psychotropic use of lithium is associated with

Select one:

- Sen and Ghose
- Bleuler
- Delay and Deniker
- Cade
- Kane

Explanation:

The use of lithium in psychiatry goes back to the mid-19 th century. Early work, however, was soon forgotten, and John Cade is credited with reintroducing lithium to psychiatry for mania in 1949.

Ref: Shorter E. The history of lithium therapy. Bipolar disorders. 2009 Jun;11:4-9.

The correct answer is: Cade

Question **15**

Not answered

Zopiclone belongs to the chemical class of

Select one:

Marked out of 1.00

Flag question

- Butyrophenone
- Cyclopyrrolones
- Imidazopyridine
- Pyrazopyrimidine
- Azaspirones

Zopiclone- cyclopyrrolone Zolpidem-imidazopyridine Zaleplon-pyrazopyrimidine

The correct answer is: Cyclopyrrolones

Question **16**

Not answered

Marked out of 1.00

Flag question

Which of the following is linked with intentional nonadherence?

Select one:

- Missed doses after long period on treatment
- High degree of trust in the treating doctor
- Low degree of self-efficacy
- Receipt of adequate levels of drug information
- Higher severity of illness

Explanation:

Patients with **poor insight** may still take medications, though lack of insight is the most common cause of poor adherence. The relationship between dose strength and adherence is probably curvilinear, with very low doses being associated with poor efficacy and very high doses with excessive side-effects. High levels of **self-efficacy** and **internal health locus of control** are consistently found to promote medication adherence. (Nafradi et al. 2017)

Factors	Reasons for intentional non-adherence
Illness-related	Lack of motivation Poor insight Grandiosity Delusions Cognitive deficit Thought disorder
Treatment-related	Side effects (weight gain, sedation, EPSEs, parkinsonism, sexual dysfunction) Dysfunctional beliefs
Clinician- and organisational-related	Therapeutic alliance Lack of follow-ups Limited consultation time
Patient-related	Denial Insight Co-morbidity Physical impairments/barriers
Environmental and cultural	Family's beliefs Cultural beliefs Religious beliefs
	Ref: Maudsley Prescribing Guidelines, 2021. p. 899; Oxford Handbook of Psychiatry, 2019. p. 994

SPMM COURSE

Ref: Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th

Edition. 2021. p. 899.

Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 994.

Nafradi L et al. Is patient empowerment the key to promote adherence? A systematic review of the relationship between self-efficacy, health locus of control and medication adherence. PloS one. 2017 Oct 17;12(10):e0186458.

The correct answer is: Low degree of self-efficacy

Question **17**

Not answered

Marked out of 1.00

Flag question

Buspirone is a

Select one:

- GABA agonist
- 5-HT1A partial agonist
- 5-HT1A agonist
- 5-HT1A antagonist
- GABA antagonist

Explanation:

Buspirone is classified as an azapirone (or azaspirone) and is chemically distinct from other psychotropic agents. It is a 5-HT1A partial agonist.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 672.

Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. p. 176.

The correct answer is: 5-HT1A partial agonist

Question **18**

Not answered

Marked out of 1.00

Flag question

Which of the following is termed as a norepinephrine and dopamine re-uptake inhibitor?

Select one:

- Varenicline
- Nefazadone
- Buspirone
- Mirtazapine
- Bupropion

Explanation:

Bupropion is a noradrenergic and dopaminergic reuptake inhibitor. It is indicated for depression but is only licensed in the UK for treatment of nicotine dependence.

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 290.

The correct answer is: Bupropion

Question **19**

Not answered

Marked out of 1.00

Flag question

A placebo is best described as which one of the following?

Select one:

- A preparation of active compound without a label
- An active preparation of precursor molecules
- A preparation with higher than average propensity to cause side effects
- A compound that is active for an unrelated disorder
- A dummy preparation of inactive compound

Explanation:

'Placebo' from latin 'placare', 'to please', entered the medical lexicon in Hooper's Medical Dictionary in 1811 as 'an epithet given to any medicine adopted to please rather than benefit the patient'. Modern study of the 'placebo effect' began when the anaesthetist Henry Beecher described patient responses to oral analgesics and later discussed 'the powerful placebo'; he found that around 30% of the clinical effect could be attributed to the effect of placebo. Placebo and nocebo effects are the effect of patients' positive and negative expectations, respectively, concerning their state of health. These effects occur in many clinical contexts, including treatment with an active agent or placebo in clinical practice or a clinical trial, the informed-consent process, the provision of information about medical treatments, and public health campaigns. Placebo effects cause beneficial outcomes, and nocebo effects cause harmful and dangerous outcomes. Placebos can be pharmacologically active or inert substances. An 'activeplacebo' has some activity, but not against the treated condition. In clinical trials, active placebos contain ingredients that cause some of the adverse effects of the active intervention; they are more difficult to distinguish from the active intervention than an inert placebo, leading to more successful blinding, less expectation bias, and a more accurate (usually smaller) estimate of apparent benefit of the active interventions.

Ref: Colloca L, Barsky AJ. Placebo and nocebo effects. New England Journal of Medicine. 2020 Feb 6;382(6):554-61.

Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. pp. 18-19.

Howick J, Hoffmann T. How placebo characteristics can influence estimates of intervention effects in trials. Cmaj. 2018 Jul 30;190(30):E908-11.

The correct answer is: A dummy preparation of inactive compound

Question **20**

Not answered

Marked out of 1.00

Flag question

Which one of the following drugs is not correctly paired with its proponents?

Select one:

- Chlorpromazine-Delay and Deniker
- Imipramine-Kuhn
- Clozapine-Bowden
- Lithium-John Cade
- Haloperidol-Janssen

Explanation:

Shortly after the introduction of clozapine to clinical practice in the mid-1970s, it was withdrawn because of several episodes of fatal agranulocytosis in patients on treatment. It was thought to have special efficacy in treatment-resistant schizophrenia, and this clinical belief was supported by an important trial by Kane et al. (1988), leading to its reintroduction in psychiatric practice, albeit with strict limitations to its prescription.

Bowden led the first randomised study on valproate (in the form of divalproex) in acutely manic patients (published in 1994).

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 218.

Bowden CL. New concepts in mood stabilization: evidence for the effectiveness of valproate and lamotrigine. Neuropsychopharmacology. 1998 Sep;19(3):194-9.

The correct answer is: Clozapine-Bowden

Question **21**

Not answered

Marked out of 1.00

Flag question

Which drug could be used for the treatment of mixed affective episodes of bipolar disorder?

Select one:

- Carbamazepine
- Gabapentin
- Sodium Valproate
- Clonazepam
- Lamotrigine

Explanation:

Mixed affective states are characterised by combinations of manic and depressive symptoms during one episode and may occur in up to 40% of bipolar patients. Mixed affective states have been insufficiently researched to arrive at definitive recommendations for treatment. Sixty six percent of these patients exhibit a poor response to lithium (SIGN Guidelines on bipolar disorder: Guideline 82 Section 3: Retrieved from http://www.sign.ac.uk/guidelines/fulltext/82/section3.html). Nevertheless, NICE recommends that mixed affective episodes are treated as manic episodes. Antipsychotics, lithium as well as sodium valproate are first line options for the treatment of acute manic episode and prophylaxis of bipolar affective disorder, and thus can be used as first line for mixed affective episodes as well.

NICE Guidelines for Bipolar Disorder (updated 2020): Managing MANIA OR HYPOMANIA in Adults in Secondary Care

- If patient is taking an **antidepressant as monotherap**y:
  1. **consider stopping the antidepressant**
  2. **offer an antipsychotic** (haloperidol, olanzapine, quetiapine, or risperidone) regardless of whether the antidepressant is stopped
- If **first antipsychotic** is **poorly tolerated or ineffective**, offer **alternative antipsychotic**
- If **alternative antipsychotic** is **not sufficiently effective** at maximum licensed dose **, consic adding lithium; if lithium** is **ineffective or not suitable, consider adding valproate** instead
- If the patient is taking an **antidepressant in combination with a mood stabilizer**, consider **stopping the antidepressant**

- If patient is **already taking lithium**, **check plasma lithium levels** to optimize treatment; conside **adding** one of the above **antipsychotics**
- If patient presents in a **mixed state** (both **manic and depressive symptoms**), follow above recommendations for **treatment of mania** and **monitor closely for depression**
- **Do not offer lamotrigine** to treat mania
- **Within 4 wks of resolution of symptoms**, discuss whether to **continue treatment for man start long-term treatment** (see below)
- If patient decides to **continue treatment for mania**, offer for **3-6 more months**, **then review**

Ref: National Institute for Health and Clinical Excellence. The NICE guideline on the assessment and management of bipolar disorder in adults, children and young people in primary and secondary care. Updated edition 2020.

The correct answer is: Sodium Valproate

Question **22**

Not answered

Marked out of 1.00

Flag question

Which of the following is an inhibitor of norepinephrine and serotonin re-uptake?

Select one:

- Reboxetine
- Bupropion
- Galantamine
- Duloxetine
- Aripiprazole

Explanation:

Duloxetine, venlafaxine, desvenlafaxine, milnacipran, and levomilnacipran are all serotonin-norepinephrine reuptake inhibitors (SNRIs).

Ref: Stahl SM. Stahl's Essential Psychopharmacology: Neuroscientific Basis and Practical Applications, 5th Edition. 2021. pp. 302-303.

The correct answer is: Duloxetine

Question **23**

Not answered

Marked out of 1.00

Flag question

Which of the following psychotropic agents has the same pharmacokinetic properties in Asians and White Europeans?

Select one:

- Alprazolam
- Haloperidol
- Diazepam
- Chlorpromazine
- Lithium

Explanation:

95% of lithium is excreted unchanged in the urine. Its excretion does not differ between Asians and White Europeans.

Ref: Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. p. 744.

The correct answer is: Lithium

Question **24**

Not answered

Marked out of 1.00

Flag question

Which one of the following was the first antidepressant to be introduced?

Select one:

- SNRI
- Tetracyclic
- SSRI
- Tricyclic
- MAO inhibitor

Explanation:

The first drug used for the treatment of depression was iproniazid in the 1950s. The choice of this drug was based on its euphoric effects in tuberculosis patients. Subsequently, it was demonstrated that iproniazid was a monoamine-oxidase (MAO) inhibitor, particularly MAO-A. The role of MAO-A is to convert biogenic (e.g. serotonin and catecholamines) and sympathomimetic (e.g. tyramine and benzylamine) amines into deaminated products. This discovery led to the development of the first class of antidepressants, the MAO inhibitors (MAOIs). Examples of this class include tranylcypromine and phenelzine.

Subsequently, the prototype antidepressant molecule, desmethylimipramine, was developed. The development of imipramine gave rise to the tricyclic antidepressants (TCAs), a class of drugs whose mechanism of action involves the inhibition of noradrenaline and serotonin reuptake by neurons. Later on, the monoamine serotonin was related to the pathophysiology of depression. Based on this discovery, the first family of psychoactive drugs with a rational design, the serotonin-specific re-uptake inhibitors (SSRIs) was developed. Thanks to the discovery and subsequent therapeutic use of MAOIs and TCAs in the 1960s, the monoaminergic theories of depression were developed.

Ref: Macedo D et al. Antidepressants, antimicrobials or both? Gut microbiota dysbiosis in depression and possible implications of the antimicrobial effects of antidepressant drugs for antidepressant effectiveness. Journal of affective disorders. 2017 Jan 15;208:22-32.

The correct answer is: MAO inhibitor

Question **25**

Not answered

Marked out of 1.00

Flag question

Which of the following is a true statement about the placebo effect?

Select one:

- Placebo response is perceived to be better for tablets than capsules
- Placebo effects do not operate on active drugs
- Prior therapeutic experiences can drive nocebo effects
- Severely depressed subjects respond better to placebos than mildly depressed ones

- Oral medications elicit a stronger placebo effect than injections

Explanation:

Placebo and nocebo effects are the effect of patients' positive and negative expectations, respectively, concerning their state of health. These effects occur in many clinical contexts, including treatment with an active agent or placebo in clinical practice or a clinical trial, the informed-consent process, the provision of information about medical treatments, and public health campaigns. Placebo effects cause beneficial outcomes, and nocebo effects cause harmful and dangerous outcomes. The molecular events and neural network changes underlying placebo and nocebo effects are mediated by expectancies or anticipated future outcomes. Prior therapeutic experiences and learning mechanisms can drive placebo and nocebo effects.

Optimism, suggestibility, and gender do NOT appear to be closely associated with placebo responsiveness. There is some evidence that among people taking active drugs, the nocebo effect is more likely to occur in those who are more anxious, have a history of medically unexplained symptoms, or have greater psychological distress and have prior therapeutic experience with the drug (cue-association).

Colour, shape and packaging also affect placebo response (capsules and larger pills have more placebo responsiveness). Placebo effects are seen for active pills as well as inert pills; thus for every prescribed drug, the clinical benefits in the real-world are obtained from both its active chemical and placebo effects.

Ref: Colloca L, Barsky AJ. Placebo and nocebo effects. New England Journal of Medicine. 2020 Feb 6;382(6):554-61.

Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. pp. 18-19.

The correct answer is: Prior therapeutic experiences can drive nocebo effects

Question **26**

Not answered

Marked out of 1.00

Flag question

Venlafaxine is best described as a

Select one:

- SNRI (Serotonin and noradrenaline reuptake inhibitor)
- SSRI (Selective serotonin reuptake inhibitor)
- NARI (Nor adrenaline reuptake inhibitor)
- Tricyclic antidepressant
- Alpha 2 adrenoreceptor antagonist

Explanation:

Duloxetine, venlafaxine, desvenlafaxine, milnacipran, and levomilnacipran are all serotonin-norepinephrine reuptake inhibitors (SNRIs).

Ref: Stahl SM. Stahl's Essential Psychopharmacology: Neuroscientific Basis and Practical Applications, 5th Edition. 2021. pp. 302-303.

The correct answer is: SNRI (Serotonin and noradrenaline reuptake inhibitor)

Question **27**

Not answered

Marked out of 1.00

Flag question

The term active placebo refers to which one of the following?

Select one:

- The placebo loses its effect with repeated use
- The placebo produces unexpected side effects
- The placebo gains positive effect with repeated use
- The placebo has activity against the treated illness
- The placebo has some activity inherently

An active placebo has some activity, but not against the treated condition. In clinical trials, active placebos contain ingredients that cause some of the adverse effects of the active intervention. They are more difficult to distinguish from the active intervention than an inert placebo, leading to more successful blinding, less expectation bias, and a more accurate (usually smaller) estimate of apparent benefit of the active interventions.

When a drug administered as placebo produces prominent side-effects, it is known as a 'nocebo'. Placebo sag is a term used to refer to decrease in placebo effect with repeated or chronic administration of placebo drugs.

Ref: Howick J, Hoffmann T. How placebo characteristics can influence estimates of intervention effects in trials. Cmaj. 2018 Jul 30;190(30):E908-11.

The correct answer is: The placebo has some activity inherently

Question **28**

Not answered

Marked out of 1.00

Flag question

Which of the following chemical classes of drugs used as antidepressants is known to increase seizure risk?

Select one:

- Triazolopyridines
- Hydrazine derivatives
- SSRIs
- Cyclopyrrolones
- Aminoketones

Explanation:

Bupropion is an aminoketone that has a dose-related risk of seizures (particularly with instant-release formulations). The risk is less with slow-release formulations at doses under 300mg/day. It is recommended to avoid the use of bupropion in people with epilepsy because it can be pro-convulsive at therapeutic doses.

Ref: Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. p. 789.

The correct answer is: Aminoketones

Question **29**

Duloxetine and milnacipran are classified as

Not answered

Marked out of 1.00

Flag question

Select one:

- Monoamine oxidase inhibitors
- Tricyclic antidepressants
- Selective serotonin reuptake inhibitors
- Serotonergic and noradrenergic reuptake inhibitors
- Adrenergic type 2 antagonists

Explanation:

Duloxetine, venlafaxine, desvenlafaxine, milnacipran, and levomilnacipran are all serotonin-norepinephrine reuptake inhibitors (SNRIs).

Ref: Stahl SM. Stahl's Essential Psychopharmacology: Neuroscientific Basis and Practical Applications, 5th Edition. 2021. pp. 302-303.

The correct answer is: Serotonergic and noradrenergic reuptake inhibitors

Question **30**

Not answered

Marked out of 1.00

Flag question

Which of the following psychotropics is well known to cause sudden death?

Select one:

- Temazepam
- Procyclidine
- Zopiclone
- Lithium
- Thioridazine

Explanation:

Several antipsychotic drugs, belonging both to typical and atypical classes (e.g. sertindole, thioridazine, and haloperidol) are known to be associated with an increased risk of QTc prolongation, potentially leading to serious ventricular arrhythmias such as torsades de pointes and sudden cardiac death. In 2005, thioridazine was withdrawn from the global market because of the risk of sudden cardiac death.

Ref: Oteri A et al. Prescribing pattern of antipsychotic drugs during the years 1996–2010: a population-based database study in Europe with a focus on torsadogenic drugs. British journal of clinical pharmacology. 2016 Aug;82(2):487-97.

The correct answer is: Thioridazine

Question **31**

Not answered

Marked out of 1.00

Flag question

Placebo effect is NOT mediated via

Select one:

- nocebo response
- natural remission
-

classical conditioning

- expectancy
- operant conditioning

Explanation:

Placebo and nocebo effects are the effect of patients' positive and negative expectations, respectively, concerning their state of health. These effects occur in many clinical contexts, including treatment with an active agent or placebo in clinical practice or a clinical trial, the informed-consent process, the provision of information about medical treatments, and public health campaigns. Placebo effects cause beneficial outcomes, and nocebo effects cause harmful and dangerous outcomes. The molecular events and neural network changes underlying placebo and nocebo effects are mediated by expectancies or anticipated future outcomes. Learning and classical conditioning play roles in both placebo and nocebo effects.

Ref: Colloca L, Barsky AJ. Placebo and nocebo effects. New England Journal of Medicine. 2020 Feb 6;382(6):554-61.

The correct answer is: nocebo response

Question **32**

Not answered

Marked out of 1.00

Flag question

An 81-year-old woman with dementia presented to the hospital with chest pain. Upon admission, she becomes acutely confused and agitated. She is combative and attacks a nurse who tried to pacify her. She refuses to accept a nurse to sit beside her and observe her continuously and threatens to kick anyone who attempts this. The best option available for the medical team is

Select one:

- Use heavy sedation to prevent physical attacks
- Discharge her back to home
- Use temporary physical or pharmacological restraints with regular re-evaluation of her medical status
- Continue to try verbal de-escalation to calm her down
- Transfer her to a psychiatric ward

Using rapid tranquillisation or physical restraints is sometimes necessary for the elderly to reduce harm to the patient and/or the health professionals. Only a minimal amount of restraint should be used, and the need for continued restraint re-evaluated frequently. Transferring to a psychiatry ward does not seem practical at this time due to the acute level of patient agitation and his behaviors, and verbal de-escalation has already failed to calm the patient. Involving a carer may help to calm the patient down, and that individual may be able to assist in making decisions for his care.

The correct answer is: Use temporary physical or pharmacological restraints with regular re-evaluation of her medical status

Question **33**

Not answered

Marked out of 1.00

Rivastigmine is a

Select one:

- Irreversible inhibitor of AchEs

Flag question

- NMDA receptor antagonist
- Reversible non-competitive inhibitor of AchEs
- Competitive inhibitor of AchEs
- Reversible competitive inhibitor of AchEs

Explanation:

Rivastigmine is a reversible and non-competitive inhibitor of acetylcholinesterase. It inhibits both acetylcholinesterase and butyrylcholinesterase.

Characteristic of cognitive enhancer	Donepezil	Rivastigmine	Galantamine	Memantine
Primary mechanism	AChE-I (selective and reversible)	AChE-I (reversible, non-competitive inhibitor)	AChE-I (competitive and reversible)	NMDA receptor antagonist (non-competitive)
Other mechanism	None	BuChE-I	Nicotine receptor agonist	5-HT3 receptor antagonist
Starting dose	5 mg daily	1.5 mg bd (oral) (or 4.6 mg/24 hours patch)	4 mg bd solution or 8 mg XL daily	5 mg daily
Usual treatment dose (and max dose)	10 mg daily	3-6 mg bd (oral) or 9.5 mg/24 hours patch	8-12 mg bd solution or 16-24 mg XL daily	20 mg daily or (10 mg bd)
Adverse effects	Diarrhoea, nausea, and headache most common	Anorexia, dizziness, nausea, vomiting, and diarrhoea most common	Nausea and vomiting most common	Dizziness, headache, fatigue, diarrhoea, and gastric pain most common
Half-life (hours)	~70 (permitting once daily dosing)	~1 (oral) – active for 10 3.4 (patch)	7-8 (oral solution) 8-10 (XL capsules)	60-100
Metabolism	CYP3A4 CYP2D6 (minor)	Minimal involvement of CYP isoenzymes	CYP3A4 CYP2D6	Primarily non-hepatic
Drug-drug interactions	Yes	Interactions unlikely	Yes	Yes
Effect of food on absorption	None	Delays rate and extent of absorption	Delays rate but not extent of absorption	None
Other	Absorbed completely from the GI tract Fairly well-tolerated and widely used	More likely to cause GI adverse effects than donepezil due to more peripheral activity	More likely to cause GI and neuropsychiatric adverse effects than donepezil	Generally better tolerated than cholinesterase inhibitors

AChE-I = acetylcholinesterase inhibitor
BuChE-I = butyrylcholinesterase inhibitor

Adapted from Maudsley Prescribing Guidelines, 2021. pp. 605-6.

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Ref: Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. pp. 605-606.

The correct answer is: Reversible non-competitive inhibitor of AchEs

Question **34**

Not answered

Marked out of 1.00

Flag question

Buspirone belongs to the chemical class of

Select one:

- Azaspirone
- Cyclopyrrolone
- Imidazopyridine
- Pyrazopyrimidine
- Butyrophenone

Explanation:

Buspirone is classified as an azapirone (or azaspirone) and is chemically distinct from other psychotropic agents. It is a 5-HT1A partial agonist.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 672.

Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. p. 176.

The correct answer is: Azaspirone

Question **35**

Not answered

Marked out of 1.00

Flag question

Which one among the following is a benzisoxazole derivative?

Select one:

- Amisulpride
- Clozapine
- Risperidone
- Sulpiride
- Ziprasidone

Explanation:

Numerous benzisoxazole moieties have been found to possess a broad spectrum of pharmacological activities. Risperidone is a 1,2-benzisoxazole ring containing drug. Sulpiride and amisulpride are substituted benzamides. Ziprasidone, rather confusingly, is a benzisothiazole, while clozapine is a dibenzodiazepine.

Ref: Rakesh KP et al. Benzisoxazole: a privileged scaffold for medicinal chemistry. MedChemComm. 2017;8(11):2023-39.

The correct answer is: Risperidone

Question **36**

Not answered

Marked out of 1.00

Flag question

Reserpine was originally introduced as an antipsychotic. It is extracted from which of the following plants?

Select one:

- Hypericum alkaloids
- Solanacea henbane
- Rauwolfia serpentina
- Hellebore
- Belladonna

Explanation:

Rauvolfia spp. (also known as Rauwolfia or devil peppers) are a group of evergreen shrubs and trees. Among the approximately 76 various species, Rauvolfia serpentina is an important medicinal plant. It is commonly known as the Indian snakeroot plant or Sarpagandha. The plant is rich in multiple secondary metabolites including reserpine, ajmaline, ajmalicine, serpentine, and yohimbine. Since ancient times, roots (mainly due to reserpine) have been utilised in various Ayurvedic and Unani medicinal preparations for the treatment of diseases like hypertension, anxiety, insomnia, and schizophrenia.

Ref: Mukherjee E et al. Biotechnological interventions on the genus Rauvolfia: recent trends and imminent prospects. Applied microbiology and biotechnology. 2019 Sep;103(18):7325-54.

The correct answer is: Rauwolfia serpentina

Question **37**

Not answered

Marked out of 1.00

Flag question

Which one of the following names is associated with the use of valproate in mania?

Select one:

- Delay and Deniker
- John Cade
- Kuhn
- Janssen
- Bowden

Explanation:

Bowden led the first randomised study on valproate (in the form of divalproex) in acutely manic patients (published in 1994). It was large in scope, was conducted at eight academic centres led by investigators with experience in bipolar disorder, included women of child-bearing potential, and included a lithium treatment group. It found divalproex to be highly statistically significant superior to placebo. Patients with mixed mania who were treated with lithium had significantly poorer responses than mixed mania patients who were treated with divalproex, or than pure manic patients treated with lithium. The response of pure manic patients did not differ significantly between the divalproex- and lithium-treated patients.

Ref: Bowden CL. New concepts in mood stabilization: evidence for the effectiveness of valproate and lamotrigine. Neuropsychopharmacology. 1998 Sep;19(3):194-9.

The correct answer is: Bowden

Question **38**

Not answered

Marked out of 1.00

Flag question

Which one of the following led a trial that proved clozapine's effectiveness in treating resistant schizophrenia?

Select one:

- Kretschmer
- Cade
- Kraepelin
- Kane
- Bleuler

Explanation:

Shortly after clozapine’s introduction to clinical practice in the mid-1970s, it was withdrawn because of several episodes of fatal agranulocytosis in patients on treatment. It was thought to have special efficacy in treatment-resistant schizophrenia, and this clinical belief was supported by an important trial by Kane et al. (1988), leading to its reintroduction in psychiatric practice, albeit with strict limitations to its prescription. 268 patients with treatment-resistant schizophrenia (failing *three* different conventional neuroleptic trials, including a six-week trial of haloperidol at maximum (60mg) doses) were randomly assigned to clozapine up to 900 mg or chlorpromazine up to 1800 mg. 30% of

the clozapine group responded while only 4% in the chlorpromazine group responded.

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 218.

The correct answer is: Kane

Question **39**

Not answered

Marked out of 1.00

Flag question

Which of the following best describes lofexidine?

Select one:

- Natural opioid
- Alpha 1 antagonist
- Alpha 2 agonist
- Synthetic opioid
- Opioid antagonist

Explanation:

Lofexidine is an alpha-2 adrenergic agonist. It can be useful to relieve symptoms of opioid withdrawal.

Ref: Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. pp. 228, 487.

The correct answer is: Alpha 2 agonist

Question **40**

Not answered

Marked out of 1.00

Flag question

Which of the following is a hormone that is used in the treatment of depression?

Select one:

- Cortisol
- Vasopressin
- Prolactin
- Thyroid hormone
- Growth hormone

Explanation:

The link between thyroid function and depression is well-studied in psycho-endocrinology. Notably, hypothyroidism can be accompanied by depression. All the hormones of the hypothalamic- pituitary-thyroid axis have been used in the treatment of depression, alone or in combination with other agents, although the most commonly used are thyroxine and levothyroxine.

Tri-iodothyronine may be used as a second-line augmentation agent in the management of treatment-resistant depression. Current evidence has shown that it has no advantage over an antidepressant alone in non-refractory illness. It may also be effective in bipolar depression.

Ref: Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. p. 321.

The correct answer is: Thyroid hormone

Question **41**

Not answered

Marked out of 1.00

Flag question

Acamprosate is a

Select one:

- Synthetic taurine analogue
- Type of benzodiazepine
- Long acting opioid agonist
- Alpha-1 agonist
- Opioid receptor antagonist

Explanation:

Acamprosate is a synthetic taurine analogue that acts as a functional glutaminergic NMDA antagonist and also increases GABA-ergic function.

Ref: Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. p. 463

The correct answer is: Synthetic taurine analogue

Question **42**

Not answered

Marked out of 1.00

Flag question

Which of the following best applies to the chemical structure of buspirone?

Select one:

- Imidazopyridine
- Benzisoxazole
- Thienobenzodiazepine
- Aminoketone
- Azaspirone-decanedione

Explanation:

Buspirone is classified as an azapirone (or azaspirone) and is chemically distinct from other psychotropic agents. It is a 5-HT1A partial agonist.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 672.

Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. p. 176.

The correct answer is: Azaspirone-decanedione

Question **43**

Not answered

Marked out of 1.00

Flag

A patient is diagnosed with mild cognitive impairment. Which one of the following is recommended for prevention of dementia?

Select one:

- Piracetam

question

- Donepezil
- Vitamin E
- Memantine
- No recommendation

Explanation:

To date, there have been no positive studies to inform prescribing in Mild Cognitive Impairment (MCI). Cochrane and other reviews show lack of efficacy of cholinesterase inhibitors, equivocal findings with piracetam, and there is no evidence to support nootropics. There have been no studies on memantine in MCI. Other studies including RCTs of vitamin E and anti-inflammatory drugs have been negative.

Ref: O'Brien et al. Clinical practice with anti-dementia drugs: a revised (third) consensus statement from the British Association for Psychopharmacology. Journal of Psychopharmacology. 2017 Feb;31(2):147-68.

The correct answer is: No recommendation

Question **44**

Not answered

Marked out of 1.00

Flag question

A potential advancement in the treatment of dementia is

Select one:

- Drugs acting on dendritic receptors
- Drugs acting on plasma enzymes
- Drugs acting on presynaptic receptors
- Drugs acting on intracellular secretase enzymes
- Drugs acting on synaptic enzymes

Explanation:

The beta-secretase BACE1 (beta-site amyloid precursor protein cleaving enzyme 1) was initially cloned and characterised in 1999. It is required for the generation of all monomeric forms of amyloid-beta, including amyloid-beta 42, which likely initiates toxicity in Alzheimer's disease (AD). BACE1 is a prime target for slowing down amyloid-beta production in early AD. Despite BACE1 inhibitor clinical trials conducted thus far being discontinued for futility or safety reasons, BACE1 remains a well-validated therapeutic target for AD.

Ref: Hampel H, et al. The β-secretase BACE1 in Alzheimer's disease. Biological psychiatry. 2021 Apr 15;89(8):745-56.

The correct answer is: Drugs acting on intracellular secretase enzymes

Question **45**

Not answered

Marked out of 1.00

Flag question

Which of the following antidepressant is an enantiomer of another antidepressant drug?

Select one:

- Fluoxetine
- Paroxetine

- Citalopram
- Escitalopram
- Sertraline

Explanation:

Citalopram is comprised of two enantiomers, R and S, which are mirror images of each other. Citalopram has mild antihistamine properties that reside in the R enantiomer and a somewhat inconsistent therapeutic action at the lowest dose, often requiring dose increase to optimise treatment. However, dose increase is limited due to the potential of QTc prolongation at higher doses. Escitalopram is comprised of only the pure active S enantiomer. This appears to remove the antihistaminic properties and there are no higher dose restrictions to avoid QTc prolongation. In addition, removal of the potentially interfering R enantiomer makes of the lowest dose of escitalopram more predictably efficacious.

Ref: Stahl SM. Stahl's Essential Psychopharmacology: Neuroscientific Basis and Practical Applications, 5th Edition. 2021. pp. 295-296.

The correct answer is: Escitalopram

Question **46**

Not answered

Marked out of 1.00

Flag question

Which one among the following is a butyrophenone derivative?

Select one:

- Chlorpromazine
- Thioridazine
- Haloperidol
- Flupenthixol
- Zuclopenthixol

Explanation:

Haloperidol, droperidol, and benperidol are butyrophenones. They are high potency, with the potential for troublesome extra-pyramidal side effects.

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 209.

Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 229.

The correct answer is: Haloperidol

Question **47**

Not answered

Marked out of 1.00

Flag question

You are giving a talk to your hospital staff about the importance of clinical trials. How would you explain the term 'standardised difference in effectiveness'?

Select one:

- The difference in the effect of treatment in the active treatment group compared to the placebo group after taking into account the variation in the treatment effect
- The effect of treatment in the active treatment group

- The difference in the effect of treatment in the active treatment group compared to the placebo group
- The number of patients that require one treatment compared to the other in order to detect one positive response
- The effect of treatment in the placebo group

Explanation:

The difference in the effect of treatment in the active treatment group compared to the placebo group after taking into account the variation in the treatment effect is the standardised measure of effectiveness. The more common term for this is 'effect size'. Effect size can be represented as a standardised difference in means (standardised mean difference) or proportions (risk difference).

The standardised mean difference (SMD) measure of effect is used when studies report efficacy in terms of a continuous measurement, such as a score on a pain-intensity rating scale. The SMD is also known as Cohen's *d*. An SMD of zero means that the new treatment and the placebo have equivalent effects.

The risk difference is often inverted (1/risk-reduction) to generate a measure called the NNT. The number of patients that require one treatment compared to the other in order to detect one positive response is the NNT (Number Needed to Treat).

Ref: Faraone SV. Interpreting estimates of treatment effects: implications for managed care. Pharmacy and Therapeutics. 2008 Dec;33(12):700.

The correct answer is: The difference in the effect of treatment in the active treatment group compared to the placebo group after taking into account the variation in the treatment effect

Question **48**

Not answered

Marked out of 1.00

Flag question

Before approval of a drug molecule, mutagenicity, carcinogenicity and organ system toxicity are studied at which phase of clinical trials?

Select one:

- Preclinical animal phase
- Phase 1
- Phase 3
- Postmarketing surveillance
- Phase 2

Explanation:

Preclinical development encompasses the activities that link drug discovery to initiation of human clinical trials. Rodent and nonmammalian models are used to delineate the pharmacokinetic profile and general safety, as well as to identify toxicity patterns. Toxicology and safety studies identify potential target organs for adverse effects and define the Therapeutic Index to set the initial starting doses in clinical trials. Mutagenicity, carcinogenicity and organ system toxicity are studied at this phase.

Ref: Steinmetz KL, Spack EG. The basics of preclinical drug development for neurodegenerative disease indications. BMC neurology. 2009 Jun;9(1):1-3.

The correct answer is: Preclinical animal phase

Question **49**

Not answered

Marked out of 1.00

Flag question

Zimeldine caused demyelination disorder on introduction, leading to its withdrawal. Which of the following class does it belong to?

Select one:

- MAO inhibitors
- Tricyclics
- Benzodiazepines
- Antipsychotics
- SSRIs

Explanation:

Zimeldine was the first of the SSRIs to reach the market. It was synthesised in 1971 and launched by the Swedish firm Astra in 1982, but subsequently withdrawn the next year because of adverse neurological effects.

Ref: Mulinari S. Divergence and convergence of commercial and scientific priorities in drug development: The case of Zelmid, the first SSRI antidepressant. Social Science & Medicine. 2015 Aug 1;138:217-24.

The correct answer is: SSRIs

Question **50**

Not answered

Marked out of 1.00

Flag question

Flupenthixol belongs to the chemical class of

Select one:

- Diphenylbutylpiperidine
- Piperidine phenothiazine
- Thioxanthene
- Piperazine phenothiazine
- Aliphatic phenothiazine

Explanation:

Flupentixol (also known as flupenthixol) and zuclopenthixol are thioxanthines. They have moderate sedative, antimuscarinic, and extra-pyramidal effects.

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 209.

The correct answer is: Thioxanthene

Question **51**

Not answered

Marked out of 1.00

Flag

The antidementia drug with an NMDA receptor antagonistic property is

Select one:

- Tacrine
- Galantamine

question

- Donepezil
- Rivastigmine
- Memantine

Explanation:

Memantine is an NMDA receptor antagonist.

Characteristic of cognitive enhancer	Donepezil	Rivastigmine	Galantamine	Memantine
Primary mechanism	AChE-I (selective and reversible)	AChE-I (reversible, non-competitive inhibitor)	AChE-I (competitive and reversible)	NMDA receptor antagonist (non-competitive)
Other mechanism	None	BuChE-I	Nicotine receptor agonist	5-HT3 receptor antagonist
Starting dose	5 mg daily	1.5 mg bd (oral) (or 4.6 mg/24 hours patch)	4 mg bd solution or 8 mg XL daily	5 mg daily
Usual treatment dose (and max dose)	10 mg daily	3-6 mg bd (oral) or 9.5 mg/24 hours patch	8-12 mg bd solution or 16-24 mg XL daily	20 mg daily or (10 mg bd)
Adverse effects	Diarrhoea, nausea, and headache most common	Anorexia, dizziness, nausea, vomiting, and diarrhoea most common	Nausea and vomiting most common	Dizziness, headache, fatigue, diarrhoea, and gastric pain most common
Half-life (hours)	~70 (permitting once daily dosing)	~1 (oral) – active for 10 3.4 (patch)	7-8 (oral solution) 8-10 (XL capsules)	60-100
Metabolism	CYP3A4 CYP2D6 (minor)	Minimal involvement of CYP isoenzymes	CYP3A4 CYP2D6	Primarily non-hepatic
Drug-drug interactions	Yes	Interactions unlikely	Yes	Yes
Effect of food on absorption	None	Delays rate and extent of absorption	Delays rate but not extent of absorption	None
Other	Absorbed completely from the GI tract Fairly well-tolerated and widely used	More likely to cause GI adverse effects than donepezil due to more peripheral activity	More likely to cause GI and neuropsychiatric adverse effects than donepezil	Generally better tolerated than cholinesterase inhibitors

AChE-I = acetylcholinesterase inhibitor
BuChE-I = butyrylcholinesterase inhibitor

Adapted from Maudsley Prescribing Guidelines, 2021. pp. 605-6.

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Ref: Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. pp. 605-606.

The correct answer is: Memantine

Question **52**

Not answered

Marked out of 1.00

Flag question

Choose a secondary amine from the below list:

Select one:

- Clomipramine
- Dothiepin
- Amitriptyline
- Amoxapine
- Imipramine

Secondary amines = desipramine, amoxapine, nortriptyline and protriptyline also duloxetine) [more potent mg to mg basis; less sedating; more noradrenergic, less antihistaminic or anticholinergic than tertiary].

The correct answer is: Amoxapine

Question **53**

Not answered

Marked out of 1.00

Flag question

Which drugs among the following is classified as a reversible MAOI?

Select one:

- Isocarboxazid
- Tranylcypromine
- Moclobemide
- Raclopride
- Phenelzine

Explanation:

Moclobemide is a reversible inhibitor of monoamine oxidase-A (RIMA). The 'cheese reaction' is less likely with RIMAs than traditional MAOIs.

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. pp. 282-283.

The correct answer is: Moclobemide

Question **54**

Not answered

Marked out of 1.00

Flag question

Which of the following ethnic differences in pharmacogenetics create natural alcohol deterrence?

Select one:

- Low cholinesterase
- Lack of aldehyde dehydrogenase
- High aldehyde dehydrogenase
- High alcohol dehydrogenase
- High CYP 3A4

Explanation:

Approximately 10% of ingested alcohol is subjected to first pass metabolism by alcohol dehydrogenase (ADH). The remainder is metabolised in the liver by ADH and CYP2E1. CYP2E1 and ADH convert alcohol to acetaldehyde. Acetaldehyde is further metabolised by aldehyde dehydrogenase to acetic acid and then to carbon dioxide and water. All the enzymes involved in the metabolism of alcohol exhibit genetic polymorphism. For example, the majority of people of north Asian origin are poor metabolisers via aldehyde dehydrogenase.

Ref: Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. p. 860.

The correct answer is: Lack of aldehyde dehydrogenase

Question **55**

Not answered

Marked out of 1.00

The antipsychotic drug paliperidone belongs to the class of

Select one:

- dibenzothiazepines
-

Flag question

benzixosazoles

- thienobenzodiazepines
- dibenzodiazepines
- benzamides

Explanation:

Paliperidone is a benzisoxazole derivative and is the major active metabolite of risperidone.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 610.

The correct answer is: benzixosazoles

Question **56**

Not answered

Marked out of 1.00

Flag question

Which of the following has evidence based support for managing treatment-resistant depression?

Select one:

- Fluoxetine and clonazepam
- Fluoxetine and buspirone
- Citalopram and valproate
- Venlafaxine and mirtazapine
- Duloxetine and fluoxetine

Explanation:

There is good literature support for the use of an SSRI or venlafaxine plus mianserin or mirtazapine in the management of treatment-resistant depression. These combinations are recommended by NICE, usually well-tolerated, and are widely used. Patients should be warned that there is a theoretical risk of serotonin syndrome. There is also a risk of blood dyscrasia with mianserin. Weight gain and sedation are common side effects of mirtazapine. The combination of mirtazapine with an SNRI is sometimes called 'California rocket fuel' because of these potentially powerful drugs blasting the patient out of the depths of depression.

Ref: Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. p. 319.

Stahl SM. Stahl's Essential Psychopharmacology: Neuroscientific Basis and Practical Applications, 5th Edition. 2021. p. 309.

The correct answer is: Venlafaxine and mirtazapine

Finish review

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